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Earth's Moon is about 1/400th the diameter of the Sun, but it is also 1/400th as far from us, making the Sun and the Moon the same size on the sky—a coincidence not shared by any other planet–moon combination in the solar system, allowing for uniquely photogenic total solar eclipses. Earth has also tidally locked the Moon, leaving it with identical periods of rotation on its axis and revolution around Earth. Wherever and whenever this happens, the locked moon shows only one face to its host planet.

Jupiter's system of moons is replete with oddballs. Io, Jupiter's closest moon, is tidally locked and structurally stressed by interactions with Jupiter and with other moons, pumping enough heat into the little orb to render molten its interior rocks; Io is the most volcanically active place in the solar system. Jupiter's moon Europa has enough H_2O that its heating mechanism—the same one at work on Io—has melted the subsurface ice, leaving a warmed ocean below. If ever there was a next-best place to look for life, it's here. (An artist coworker of mine once asked whether alien life forms from Europa would be called Europeans. The absence of any other plausible answer forced me to say yes.)

Pluto's largest moon, Charon, is so big and close to Pluto that Pluto and Charon have each tidally locked the other: their rotation periods and their periods of revolution are identical. We call this a "double tidal lock," which sounds like a yet-to-be-invented wrestling hold.

By convention, moons are named for Greek personalities in the life of the Greek counterpart to the Roman god after whom the planet itself was named. The classical gods led complicated social lives, so there is no shortage of characters from which to draw. The lone exception to this rule applies to the moons of Uranus, which are named for assorted protagonists in British lit. The English astronomer Sir William Herschel was the first person to discover a planet beyond those easily visible to the naked eye, and he was ready to name it after the King, under whom he faithfully served. Had Sir William succeeded, the planet list would read: Mercury, Venus, Earth, Mars, Jupiter, Saturn, and George. Fortunately, clearer heads prevailed and the classical name Uranus was adopted some years later. But his original suggestion to name the moons after characters in William Shakespeare's plays and Alexander Pope's poems remains the tradition to this day. Among its twenty-seven moons we find Ariel, Cordelia, Desdemona, Juliet, Ophelia, Portia, Puck, Umbriel, and Miranda.